LESSON 5: BOUNDED OPERATORS ON A HILBERT SPACE AND THE CONTINUOUS FUNCTIONAL CALCULUS

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In this lecture, we introduce the notion of adjoint of a bounded linear operator in a Hilbert space, and we study the main properties of the map that associates to $A \in B(H)$ the adjoint (denoted by A^*). With the notion of adjoint, we define normal operators, self-adjoint operators, and unitary operators. Then we will concentrate on the continuous functional calculus for self-adjoint operators.

1. The adjoint operator

In order to define the adjoint of a linear bounded operator on a Hilbert space, we have to recall the definition of sesquilinear forms and the theorem of representation of sesquilinear forms.

Definition 1.1 (Sesquilinear forms)**.** *Let H be a Hilbert space. A hermitian for (or sesquilinear form) is a map*

$$F : H \times H \to \mathbb{C}$$

such the for all $x, y, z \in H$ and $\lambda \in \mathbb{C}$ we have

- $F(x + y, z) = F(x, z) + F(y, z)$
- $F(x, z + w) = F(x, z) + F(x, w)$
- $F(\lambda x, y) = \lambda F(x, y)$
- $F(x, \lambda y) = \overline{\lambda} F(x, y)$

Example on the slides

Definition 1.2 (Bounded Sesquilinear forms)**.** *We say that the sesquilinear form $F : H \times H \to \mathbb{C}$ is bounded if there exists M such that*

$$|F(x, y)| \leq M\|x\|\|y\|, \qquad \forall x, y \in H.$$

Theorem 1.3 (Representation of sesquilinear forms)**.** *Let $F : H \times H \to \mathbb{C}$ be a bounded sesquilinear form. Then there exists a unique operator $B \in \mathcal{B}(H)$ such that*

$$F(x, y) = (x, By), \qquad \forall x, y \in H$$

with

$$\|B\| = \sup_{\|x\|=\|y\|=1} |F(x, y)| = M.$$

The Adjoint of a bounded operator is a consequence of the representation of sesquilinear forms.

Definition 1.4 (The adjoint A^* of a bounded linear operator A)**.** *Let $A \in \mathcal{B}(H)$. Then, the bounded sesquilinear form defined by*

$$F(x, y) := (Ax, y), \quad \forall x, y \in H,$$

that is

$$|F(x, y)| = |(Ax, y)\| \leq \|Ax\|\|y\| \leq \|A\|\|x\|\|y\|, \quad \forall x, y \in H$$

by the representation theorem of sesquilinear forms can be written in terms of a unique bounded linear operator, that we denote by A^*, as

$$F(x,y) := (Ax,y) = (x,A^*y), \quad \forall x,y \in H,$$

with

$$\|A\| = \|A^*\|.$$

The norm is the same because it's bounded

We study the properties of the involution $A \to A^*$. An important fact that will be used often is the following theorem, of which we omit the proof, and its corollary.

Theorem 1.5. *Let $A \in \mathcal{B}(H)$. If $(Ax,x) = 0$ for all $x \in H$ then $A = 0$.*

Corollary 1.6. *Let $A, B \in \mathcal{B}(H)$. If $(Ax,x) = (Bx,x)$ for all $x \in H$ then $A = B$.*

Theorem 1.7 (Properties of $A \to A^*$)**.** *Let $A \in \mathcal{B}(H)$. Then the map $A \to A^*$ from $\mathcal{B}(H)$ into itself has the properties*

(I) $(A+B)^* = A^* + B^*, \quad A, B \in \mathcal{B}(H)$
(II) $(\lambda A)^* = \overline{\lambda} A^*, \quad A \in \mathcal{B}(H), \quad \lambda \in \mathbb{C}$
(III) $(AB)^* = B^* A^*, \quad A, B \in \mathcal{B}(H)$
(IV) $A^{**} = A, \quad A \in \mathcal{B}(H)$.

Define T as the map T: B(H) --> B(H), then T(A)=A

Moreover, we have the equality

$$\|A^*A\| = \|A\|^2.$$

Proof. We prove the various steps.

(I) Property: $(A+B)^* = A^* + B^*, \quad A, B \in \mathcal{B}(H)$ follows from

$$((A+B)x,y) = (Ax,y) + (Bx,y)$$
$$= (x,A^*y) + (x,B^*y) = (x,A^*y + B^*y) = (x,(A^*+B^*)y)$$

so

$$((A+B)x,y) = (x,(A+B)^*y) = (x,(A^*+B^*)y)$$

(II) Property$(\lambda A)^* = \overline{\lambda} A^*$ follows from

$$(\lambda Ax,y) = \lambda(Ax,y) = \lambda(x,A^*y) = (x,\overline{\lambda} A^*y)$$

(III) Property $(AB)^* = B^*A^*$ follows from

$$(ABx,y) = (Bx,A^*y) = (x,B^*A^*y), \quad \text{and} \quad (ABx,y) = (x,(AB)^*y)$$

(IV) Property$(A)^{**} = A$ follows from

$$(Ax,y) = (x,A^*y) = \overline{(A^*y,x)} = \overline{(y,(A^*)^*x)} = ((A^*)^*x,y).$$

Finally the equality $\|A^*A\| = \|A\|^2$ follows from observing that

$$\|Ax\|^2 = (Ax,Ax) = (A^*Ax,x) \le \|A^*A\|\|x\|^2$$

for every $x \in H$,

$$\|A\|^2 = \sup \frac{\|Ax\|^2}{\|x\|^2} \le \|A^*A\|.$$

On the other hand, since $\|A\| = \|A^*\|$ we get

$$\|A^*A\| \leq \|A^*\|\|A\| = \|A\|^2.$$

Hence we have the equality

$$\|A\|^2 = \|A^*A\|$$

holds for every $A \in \mathcal{B}(H)$. $\square$

Definition 1.8 (Orthogonal complement)**.** *We have the following definitions.*

- *If $(x, y) = 0$ we say that x is orthogonal to y and we write $x \perp y$. The relation is obviously symmetric.*
- *If $E \subset H$ and $F \subset H$ the symbol $E \perp F$ denotes that $(x, y) = 0$ for all $x \in F$ and $y \in E$.*
- *Let $M \subset H$ be a subspace of H we denote by $M^\perp$ the orthogonal complement of M defined as*

$$M^\perp := \{x \in H \quad | \quad (x, m) = 0 \quad \forall m \in M\}.$$

Theorem 1.9. *Let $M \subset H$ be a closed subspace of H. Then*

$$H = M \oplus M^\perp$$

where the space $M^\perp$ is the orthogonal complement of M.

Remark 1.10 (Recall: Kernel and Range)**.** *Let $A : X \to Y$ be a linear operator from X to a vector space Y. The*

- *kernel (or null space) of A, denoted by $\ker(A)$, is the linear subspace:*

$$\ker(A) = \{x \in X : A(x) = 0_Y\}$$

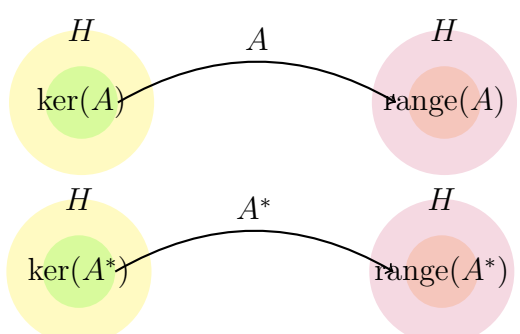
  where 0_Y is the zero vector in Y.
- *The range, or image, of A, denoted by range(A) or Im(A), is the linear subspace:*

$$\textit{range}(A) = \{A(x) \in Y : x \in X\}.$$

Theorem 1.11 (Relations among $\ker(A)$, $\ker(A^*)$, $\text{range}(A)$ and $\text{range}(A^*)$)**.** *Let $A \in \mathcal{B}(H)$, then we have*

$$\ker(A^*) = \text{range}(A)^\perp \quad \text{and} \quad \ker(A) = \text{range}(A^*)^\perp.$$

Visualization of the theorem:



Proof of the relation $\ker(A^*) = \text{range}(A)^\perp$. We recall that

$$\ker(A^*) = \{y \in H \ : \ A^*y = 0\}$$

and

$$A^*y = 0 \quad \text{if and only if} \quad (x, A^*y) = 0 \ \forall x \in H$$

so by the definition of adjoint

$$0 = (x, A^*y) = (Ax, y), \quad \forall x \in H$$

equivalent to

$$(Ax, y) = 0, \quad \forall x \in H$$

equivalent to

$$y \perp range(A).$$

this proves

$$\ker(A^*) = \text{range}(A)^\perp.$$

Now exchange the role of A and A^* and we get the second relation

$$\ker(A) = \text{range}(A^*)^\perp.$$

□

1.1. **Normal, self-adjoint and unitary operators.** In the setting of Hilbert spaces, there are important classes of operators defined below.

Definition 1.12 (normal, self-adjoint, unitary, projector)**.** *An operator $A \in \mathcal{B}(H)$ is said to be*

(i) *normal if $AA^* = A^*A$,*
(ii) *self-adjoint or hermitian if $A = A^*$,*
(iii) *unitary if $A^*A = AA^* = I$,*
(iv) *a projection if $A = A^2$.*

Remark 1.13. *Observe that self-adjoint operators and unitary operators are normal.*

Theorem 1.14 (for normal operators)**.** *Suppose $A \in \mathcal{B}(H)$:*

(a) *A is normal if and only if $\|Ax\| = \|A^*x\|$ for every $x \in H$.*
(b) *If A is normal then $\ker(A) = \ker(A^*) = \text{range}(A)^\perp$*
(c) *If A is normal and $Ax = \lambda x$ for some $x \in H$ then $A^*x = \overline{\lambda}x$*
(d) *If A is normal and λ and μ are distinct eigenvalues of A then the corresponding eigenspaces are orthogonal to each other.*

Remark 1.15. *Since self-adjoint operators are normal if we set $A^* = A$ in the relations of point (c), we have*

$$\overline{\lambda} = \lambda$$

which implies $\lambda \in \mathbb{R}$.

Theorem 1.16 (for unitary operators)**.** *Let $U \in \mathcal{B}(H)$. Then the following statements are equivalent:*

(a) *U is unitary.*
(b) $\text{range}(U) = H$ *and* $(Ux, Uy) = (x, y) \quad \forall x, y \in H$.
(c) $\text{range}(U) = H$ *and* $\|Ux\| = \|x\| \quad \forall x \in H$.

Remark 1.17. *Observe that unitary operators U are those who satisfy $\|Ux\| = \|x\|$. This means that a unitary evolution operator $U(t)$ preserves the norm. This is important for the Schrödinger evolution operator $U(t) = e^{-itH/\hbar}$ of quantum states ψ_0, the evolution preserves the probability $\|e^{-itH/\hbar}\psi_0\|_{L^2} = \|\psi(t)\|_{L^2}$.*

Theorem 1.18 (Properties of a projection $P \in \mathcal{B}(H)$)**.** *Each one of the following four properties of a projection $P \in \mathcal{B}(H)$ implies the other three:*

(i) *P is self-adjoint,*
(ii) *P is normal,*
(iii) $\text{range}(P) = \ker(P)^{\perp}$
(iv) $(Px, x) = \|Px\|^2, \quad \forall x \in H$.

Remark 1.19. *Property (i) is usually expressed by saying that P is an orthogonal projection.*

2. Approximation of continuous functions

Approximation of continuous functions with polynomials.

Theorem 2.1 (Weierstrass theorem)**.** *Let K be a compact set in $\mathbb{R}$. Then for every continuous function $f : K \to \mathbb{C}$ there exists a sequence of polynomials P_n such that $P_n \to f$ uniformly on K.*

Explicit approximation of continuous functions are given by the Bernstein's Theorem.

Theorem 2.2 (Bernstein's theorem)**.** *For every function $f(t) \in C^0[0, 1]$, the sequence of Bernstein polynomials:*

$$B_n(f, t) = \sum_{i=0}^{n} f\left(\frac{i}{n}\right)\binom{n}{i} t^i (1-t)^{n-i}$$

converges uniformly to $f(t)$, where $\binom{n}{i}$ is Newton's binomial coefficients.

Remark 2.3. *Change of Variable:*

- *A simple change of variable $t = \frac{x-a}{b-a}$ transforms $[a, b]$ into $[0, 1]$.*
- *This allows us to pass from $f(x) \in C^0[a, b]$ to $f(t) \in C^0[0, 1]$.*

Remark 2.4. *The Weierstrass function is a classic example of a function that is continuous everywhere but differentiable nowhere.*
It is defined as:

$$W(x) = \sum_{n=0}^{\infty} a^n \cos(b^n \pi x)$$

where $0 < a < 1$ and b is a positive odd integer such that $ab > 1 + \frac{3\pi}{2}$. See the plot at the end of these notes.

3. The continuous functional calculus for bounded self-adjoint operators

For self-adjoint operators (or more generally for normal operators), we can define the continuous functional calculus thanks to the Weierstrass approximation theorem.

We start with the properties of the polynomial functional calculus, and by density, we will extend the calculus to continuous functions.

Remark 3.1. *We point out that if we consider functions that are merely continuous, we must restrict the class of operators for which we define the functional calculus. This is a general principle in operator theory. If we enlarge the class of functions we consider in order to define a certain functional calculus, we must reduce the class of operators.*

Remark 3.2 (Strategy for the construction of the continuous functional calculus)**.**

- *We recall that for operators in $\mathcal{B}(H)$ the spectrum is compact and nonempty.*
- *For self-adjoint operators $\sigma(A)$ is real.*
- *We define the functional calculus for polynomials $P_n(A)$ where $A \in \mathcal{B}(H)$ and is self-adjoint.*
- *We show that $\|P_n(A)\|_{\mathcal{B}(H)} = \sup_{t\in\sigma(A)} |P_n(t)| := \|P_n\|_\infty$.*
- *We use the Weierstrass theorem to define $f(A)$ for every continuous function $f : \sigma(A) \to \mathbb{C}$ since there exists a sequence of polynomials P_n such that $P_n \to f$ uniformly on $\sigma(A)$.*

Theorem 3.3. *Let $P_n(t) = a_n t^n + \ldots + a_0$ with $a_n \neq 0$ where $a_j \in \mathbb{C}$ for $j = 0, ..., n$. We define the polynomial functional calculus as*

$$P_n(A) = a_n A^n + \ldots + a_0 I$$

where $A \in \mathcal{B}(H)$ and A is self-adjoint. Then, we have

$$\|P_n(A)\|_{\mathcal{B}(H)} = \sup_{t\in\sigma(A)} |P_n(t)|.$$

Remark 3.4. *In the following we set*

$$\|P_n\|_\infty := \sup_{t\in\sigma(A)} |P_n(t)|$$

and we write $\|P_n(A)\|$ instead of $\|P_n(A)\|_{\mathcal{B}(H)}$.

Proof. The proof of the theorem is based on the facts:

(I) The spectral mapping theorem $\sigma(P_n(A)) = P_n(\sigma(A))$ which we have already proved. Precisely

$$\sigma(P_n(A)) = \{\lambda \in \mathbb{C} \ : \ \lambda = P_n(t) \ : \ t \in \sigma(A)\}.$$

(II) Since $P_n(t) = a_n t^n + \ldots + a_0$ with $t \in \mathbb{R}$ and $a_n \neq 0$ where $a_j \in \mathbb{C}$ for $j = 0, ..., n$ we have

$$\overline{P_n}(t) = \overline{a_n} t^n + \ldots + \overline{a_0}.$$

So we have

$$|P_n(t)|^2 = P_n(t)\overline{P_n}(t)$$

and

$$\overline{P_n}(A) = \overline{a_n}A^n + \ldots + \overline{a_0}I = P_n(A)^* \quad \text{(recall that } \ (\lambda A)^* = \overline{\lambda}A^*).$$

(III) From (I) (The spectral mapping theorem) applied to the polynomial $P_n(t)\overline{P_n}(t)$ we have

$$\sigma(P_n(A)\overline{P_n}(A)) = \{|\lambda|^2 = \lambda\overline{\lambda} \ : \ \lambda\overline{\lambda} = P_n(t)\overline{P_n}(t) \ : \ t \in \sigma(A)\}$$

(IV) Now observe that

$$\|P_n\|_\infty^2 := \sup_{t\in\sigma(A)} |P_n(t)|^2 = \sup_{t\in\sigma(A)} P_n(t)\overline{P_n}(t).$$

(V) Recall that the spectral radius is defined

$$r(A) := \sup_{\lambda\in\sigma(A)} |\lambda|$$

and Gelfand formula

$$r(A) = \lim_{n\to\infty} \|A^n\|^{1/n},$$

for self-adjoint operators $A = A^*$ and thanks to the relation

$$\|A^*A\| = \|A\|^2,$$

gives

$$\|A^2\| = \|A\|^2, \quad \|A^4\| = \|A\|^4, ..., \|A^{2^k}\| = \|A\|^{2^k}....$$

so for $n = 2^k$ we get

$$\lim_{n\to\infty} \|A^n\|^{1/n} = \|A\| = r(A).$$

(VI) Since A is self-adjoint we have

$$P_n(A)\overline{P_n}(A) = P_n(A)P_n(A)^*$$

and so $P_n(A)\overline{P_n}(A)$ is self-adjoint as well.

(VII) Set

$$T := P_n(A)\overline{P_n}(A).$$

We now use

$$r(T) := \sup_{\lambda\in\sigma(T)} |\lambda|$$

and

$$r(T) = \|T\|$$

and

$$\|A^*A\| = \|A\|^2.$$

So we can write

$$\sup_{\lambda\in\sigma(P_n(A)\overline{P_n}(A)} |\lambda|^2 = r(P_n(A)P_n(A)^*) = \|P_n(A)P_n(A)^*\| = \|P_n(A)\|^2,$$

since

$$\sigma(P(A)) = \{\lambda \ : \ \lambda = P(t) \in \mathbb{C} \ : \ t \in \sigma(A)\}$$

we get

$$\|P_n(A)\|^2 = \sup_{\lambda\in\sigma(P_n(A)\overline{P_n(A)})} |\lambda|^2 = \sup_{t\in\sigma(A)} |P_n(t)|^2 := \|P\|_\infty^2.$$

This concludes the proof. □

Using the Weierstrass approximation theorem for continuous functions we now define the continuous functional calculus

$$f \to \phi(f) := f(A),$$

for bounded self-adjoint operators A on H. For every f, $g \in C(\sigma(A))$ and λ, $\mu \in \mathbb{C}$, we show that $\phi(f)$ is a $*$-homomorphism.

Definition 3.5 (of $*$-homomorphism)**.** *We say that the map*

$$f \to \phi(f) := f(A),$$

for bounded self-adjoint operators A on H, for every f, $g \in C(\sigma(A))$ and λ, $\mu \in \mathbb{C}$, is a $*$-homomorphism if it satisfies the conditions

- $\phi(\lambda f + \mu g) = \lambda\phi(f) + \mu\phi(g)$,
- $\phi(fg) = \phi(f)\phi(g)$,
- $\phi(1) = I$,
- $\phi(\overline{f}) = \phi(f)^*$.

Remark 3.6. *The notation $\phi(f)$, for the functional calculus $f(A)$, is sometimes preferable when we want to stress that we fix the operator A and we vary the function f.*

Theorem 3.7 (Uniqueness of the continuous functional calculus and properties)**.** *Let $A \in \mathcal{B}(H)$ be self-adjoint on the complex separable Hilbert space H. Then there exists a unique function*

$$\phi : C(\sigma(A)) \to \mathcal{B}(H)$$

with the properties:

(I) *ϕ is a $*$-homomorphism*
(II) $\|\phi(f)\|_{\mathcal{B}(H)} = \|f\|_\infty$ (where $\|f\|_\infty := \sup_{t\in\sigma(A)} |f(t)|$)
(III) *If $f(t) = t$ then we have $\phi(f(t)) = A$*
(IV) *If $f(t) = (\mu - t)^{-1}$, for $\mu \notin \mathbb{R}$, then we have $\phi(f) = (\mu I - A)^{-1}$.*

Proof. We divide the proof in steps.

Step 1. Definition of the continuous functional calculus $\phi : C(\sigma(A)) \to \mathcal{B}(H)$:

- Since $f \in C(\sigma(A))$ and $\sigma(A)$ is compact by the Weierstrass approximation theorem of continuous functions we can approximate f with a sequence of polynomials P_n such that

$$\|f - P_n\|_\infty = \sup_{t\in\sigma(A)} |f(t) - P_n(t)|.$$

- The natural definition of $\phi : C(\sigma(A)) \to \mathcal{B}(H)$ is then
$$\phi(f) := \lim_{n\to\infty} \phi(P_n) = \lim_{n\to\infty} P_n(A).$$

- Observe that by the spectral mapping theorem previously proved for polynomials
$$\|P_n(A)\|_{\mathcal{B}(H)} = \|P_n\|_\infty := \sup_{t\in[a,b]} |P_n(t)|$$
we have that from n, $m \in \mathbb{N}$
$$\begin{aligned}\|\phi(P_n) - \phi(P_m)\|_{\mathcal{B}(H)} &= \|\phi(P_n - P_m)\|_{\mathcal{B}(H)} \\ &= \|P_n - P_m\|_\infty := \sup_{t\in\sigma(A)} |P_n(t) - P_m(t)|\end{aligned}$$
due to the linearity of the polynomial functional calculus.

Step 2. The calculus $\phi(f) := \lim_{n\to\infty} \phi(P_n)$ is well defined:
So from the equality
$$\|\phi(P_n) - \phi(P_m)\|_{\mathcal{B}(H)} = \sup_{t\in\sigma(A)} |P_n(t) - P_m(t)| \to 0 \quad n, m \to \infty$$
it follows that

- The sequence of bounded linear operators $\phi(P_n)$ is a Cauchy sequence and it converges to an operator in $\mathcal{B}(H)$ because $\mathcal{B}(H)$ is a Banach space.
- The limit does not depend from the approximating sequence since if P_n and Q_n are such that $P_n \to f$ and $Q_n \to f$ then
$$\|\phi(P_n) - \phi(Q_n)\|_{\mathcal{B}(H)} = \sup_{t\in\sigma(A)} |P_n(t) - Q_n(t)| \to 0 \quad n \to \infty$$
- The above facts show that the functional calculus is well defined.

Step 3. Properties of the continuous functional calculus:

- Point (I): ϕ is a $*$-homomorphism follows from the fact that the polynomial functional calculus is a $*$-homomorphism. Using the approximating sequences $P_n \to f$ and $Q_n \to g$ we get the statement. For example
$$\begin{aligned}\phi(\lambda f + \mu g) &:= \lim_{n\to\infty} \phi(\lambda P_n + \mu Q_n) \\ &= \lambda \lim_{n\to\infty} \phi(P_n) + \mu \lim_{n\to\infty} \phi(Q_n) \\ &= \lambda\phi(f) + \mu\phi(g)\end{aligned}$$
- Point (II) $\|\phi\|_{\mathcal{B}(H)} = \|f\|_\infty$ follows by density
$$\|\phi(f)\|_{\mathcal{B}(H)} = \lim_{n\to\infty} \|\phi(P_n)\|_{\mathcal{B}(H)} = \lim_{n\to\infty} \|P_n\|_\infty = \|f\|_\infty.$$
- Point (III) If $f(t) = t$ we have $\phi(f(t)) = A$ is trivial it is the polynomila functional calculus.

We consider the facts:

- Point (IV) If $f(t) = (\mu - t)^{-1}$, for $\mu \notin \mathbb{R}$, then we have $\phi(f) = (\mu I - A) - 1$. Follows from the product rule: set $g(t) = (\mu - t)$, since
$$f(t)g(t) = g(t)f(t) = 1$$
by the homomorphism property
$$\phi(f)(\mu I - A) = (\mu I - A)\phi(f) = \phi(1) = I$$
so $\phi(f) = (\mu I - A)^{-1}$.

Step 4. The map $\phi : C(\sigma(A)) \to \mathcal{B}(H)$ is unique

Remark: We will prove that the map $\phi : C(\sigma(A)) \to \mathcal{B}(H)$ that satisfies the point (I)-(II) and (III) is unique.

- Let us suppose that there exists a second map $\psi : C(\sigma(A)) \to \mathcal{B}(H)$.
- If we assume that $\phi \neq \psi$ we get a contradiction.
- By the $*$-homomorphism property $\phi(1) = \psi(1) = I$
- and also for $f(t) = t$ we have $\phi(f(t)) = \psi(f(t)) = A$
- Consider the polynomial $P_n(t) = a_n t^n + \ldots + a_0$ with $a_n \neq 0$ where $a_j \in \mathbb{C}$ for $j = 0, ..., n$ and for $f(t) = t$ we have
$$P_n(t) = a_n (f(t))^n + \cdots + a_0 I = a_n f(t) \cdots f(t) + \cdots + a_0 I$$
where $f(t) \cdots f(t)$ are multiplied n-times.

We consider the facts:

- Now apply the homomorphism ψ to $P_n(t)$ and from the homomorphism property
$$\begin{aligned}
\psi(P_n(t)) &= \psi(a_n f(t) \cdots f(t) + \cdots + a_0 1) \\
&= a_n \psi(f(t) \cdots f(t)) + \cdots + a_0 \psi(1)) \\
&= a_n \psi(f(t)) \cdots \psi(f(t)) + \cdots + a_0 \psi(1) \\
&= a_n A \cdots A + \cdots + a_0 I \\
&= a_n A^n + \cdots + a_0 I \\
&= \phi(P_n)
\end{aligned}$$
- So if we assume that they are different we find a contradiction.
- Reasoning by density we proved that
$$\psi(f) = \phi(f), \quad \forall f \in C(\sigma(A)).$$

□

4. Some consequences of the continuous functional calculus

The square root of a non-negative operator is an important application of the continuous functional calculus. We will state some results without proofs.

Definition 4.1. *We say that a self-adjoint operator in $\mathcal{B}(H)$ is nonnegative if $(Ax, x) \geq 0$, $\forall x \in H$ and we write $A \geq 0$.*

Theorem 4.2. *Let A be a self-adjoint operator in $\mathcal{B}(H)$ and let $f \to \phi(f) = f(A)$ be its continuous functional calculus. Then we have*

$$f \geq 0 \Rightarrow \phi(f) = f(A) \geq 0.$$

Theorem 4.3. *Let A be a self-adjoint operator in $\mathcal{B}(H)$ then*

$$A \geq 0 \Leftrightarrow \sigma(A) \geq 0$$

The symbol $\sigma(A) \geq 0$ means that the spectrum is nonnegative.

Theorem 4.4. *Theorem (The square root of a positive operator) Let A be a self-adjoint operator in $\mathcal{B}(H)$ such that $A \geq 0$. Then there exists a unique self-adjoint and nonnegative operator B such that $B^2 = A$.*

We consider the facts:

- Given two self-adjoint operator A, $B \in \mathcal{B}(H)$ we say that $A \geq B$ if $A - B \geq 0$. This relation define a partial order, since
- $A \geq A$
- $A \geq B$ and $B \geq A \Rightarrow A = B$
- $A \geq B$ and $B \geq C \Rightarrow A \geq C$.
- It can happen that we do not have $A \geq B$ nor $B \geq A$.

Theorem 4.5. *Let $(A_n)_{n\in\mathbb{N}}$ and B be a sequence of bounded self-adjoint operators such that*

$$A_1 \leq A_2 \leq A_3 \leq \cdots \leq A_n \cdots \leq B$$

Then the sequens $(A_n)_{n\in\mathbb{N}}$ strongly converge to a bounded-self-adjoint operator $A \leq B$.

Strong convergence means that $A_n x \to Ax$ for all $x \in H$.

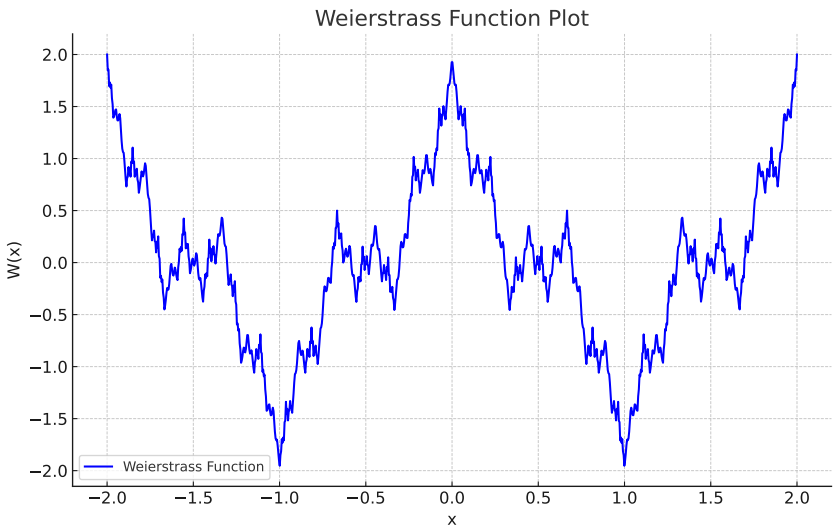


FIGURE 1. Weierstrass Function Plot